

Methods

Frontal-Temporal Functional Connectivity

Inter-regional functional connectivity between the left frontal and left temporal cortices was assessed using autoregressive Pearson correlation (ar_corr) as implemented in the NIRS Toolbox (conectividade_funcional.m). Connectivity was computed separately for each task block within each session (divide_events = true, min_event_duration = 10 s), with short-separation regressors included to reduce physiological noise (AddShortSepRegressors = 1). The correlation function used an autoregressive model order equal to four times the sampling frequency (ar_corr(data, "4xFs", true)), yielding Pearson r values, Fisher-z transformations ($Z = \text{arctanh}(r)$), and associated p and q (FDR-corrected) statistics for all channel pairs.

For inter-ROI connectivity analyses, the ROI definitions were: Frontal (6 channels: S1-D1, S2-D1, S2-D3, S1-D2, S3-D3, S3-D1) and Temporal (10 channels: S8-D7, S8-D4, S7-D7, S7-D4, S7-D6, S6-D5, S6-D4, S5-D5, S5-D6, S5-D4), yielding $6 \times 10 = 60$ inter-ROI channel pairs per session. Mean inter-ROI connectivity (conexao_inter) was computed as the average Fisher-z value across all 60 pairs for each participant-session combination.

Connectivity LMM

A LMM was estimated to test for differential changes in frontal-temporal connectivity across training sessions. Calibration session (session 0) connectivity was used as a per-participant covariate (conn_baseline) rather than a modelled time point. Three participant-session observations were excluded due to poor signal quality (SUBJ_001, SUBJ_005, and SUBJ_024 at session 4), leaving $N = 121$ observations. The model formula was:

$$\text{conexao_inter} \sim \text{group} * \text{session} + \text{conn_baseline} + (1 \mid \text{subject})$$

Model estimation used REML with the BFGS optimiser. The group x session interaction term was the critical test of whether the Accelerated group showed a steeper increase in frontal-temporal connectivity across training.

Results

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Mean inter-ROI connectivity (Fisher-z, back-transformed to r for description) across training sessions was broadly comparable between groups: Accelerated $M = 0.296$ ($SD = 0.190$) and Non-Accelerated $M = 0.255$ ($SD = 0.149$). The LMM revealed no significant group x session interaction ($\beta = 0.002$, $SE = 0.013$, $z = 0.178$, $p = .859$, 95% CI $[-0.024, 0.028]$; Table 4). No other fixed effects reached significance. Mann-

Whitney comparison of mean per-subject connectivity across training sessions found no significant group difference ($U = 33$, $p = .318$, $r = -0.347$, medium effect, n.s.).

Table 4. LMM Results — Frontal-Temporal Connectivity (HbO, Fisher-z, N = 121)

Parameter	Beta	SE	z	p	95% CI
Intercept	0.190	0.086	2.208	.027	[0.021, 0.358]
Group (Accelerated)	0.039	0.112	0.345	.730	[-0.182, 0.259]
Session	0.016	0.009	1.742	.082	[-0.002, 0.034]
Group x Session	0.002	0.013	0.178	.859	[-0.024, 0.028]
Connectivity Baseline	0.031	0.216	0.142	.887	[-0.393, 0.455]

REML; BFGS; 3 obs excluded (SUBJ_001, SUBJ_005, SUBJ_024 at session 4).